

1. 2013Q3

Question 3 refers to the table below, which shows the relative amounts of mitochondria, chloroplasts and endoplasmic reticulum in four cell types P, Q, R and S:

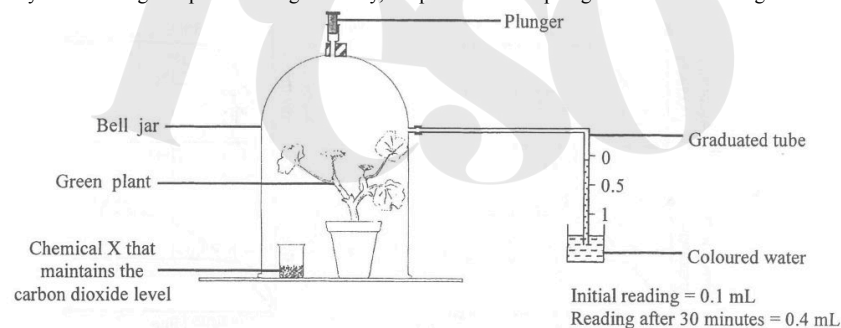
Cell type	Relative amount of the sub-cellular structure		
	Mitochondrion	Chloroplast	Endoplasmic reticulum
P	+	++	+
Q	+++	-	+
R	+++	-	+++
S	+	-	+

Which of the following carbohydrate(s) is/are likely to be found in cell type P?

- (1) starch
(2) glucose
(3) glycogen
- A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

2. 2014Q6

Questions 6 to 8 refer to the diagram below, which shows a set-up used to determine the rate of photosynthesis of a green plant. During the study, the position of the plunger remained unchanged.



Based on the results, what was the rate of photosynthesis of this plant?

- A. 0.6 mL oxygen released per hour
B. 0.3 mL oxygen released per hour
C. 0.6 mL carbon dioxide absorbed per hour
D. 0.3 mL carbon dioxide absorbed per hour

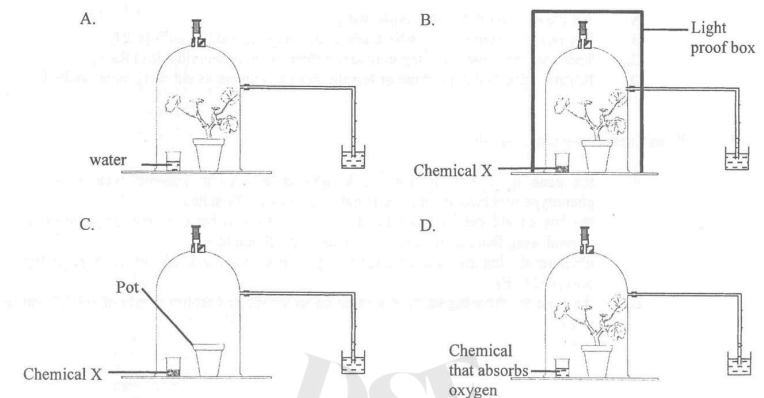
3. 2014Q7

The rate obtained was lower than the actual rate of photosynthesis of the plant. Which of the following is the most probable reason for this?

- A. The plant also carried out respiration during the study.
B. The plant also carried out transpiration during the study.
C. The air temperature might have increased during the study.
D. The atmospheric pressure might have decreased during the study.

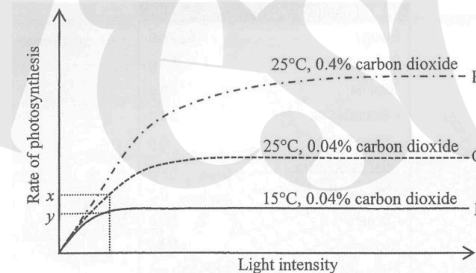
4. 2014Q8

Which of the following set-ups can be used as a control for the above study to find out the actual rate of photosynthesis?



5. 2016Q9

Questions 9 and 10 refer to the diagram below, which shows the rate of photosynthesis of a plant under different conditions:



The factor(s) that determine(s) the difference in the values of x and y in the diagram is / are

- A. temperature.
B. light intensity.
C. carbon dioxide concentration.
D. light intensity and temperature.

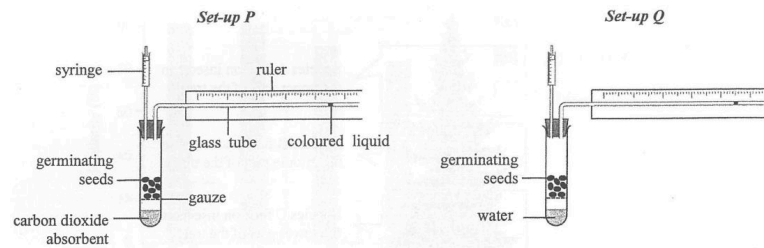
6. 2016Q10

Which of the following descriptions best accounts for the higher rate of photosynthesis of P as compared to Q?

- A. All conditions are optimum in P.
B. There is a faster diffusion of carbon dioxide into the leaves.
C. More carbon dioxide can be used as raw material for photosynthesis.
D. Carbon dioxide concentration in P is much higher than that in the atmosphere.

7. 2022Q11

Questions 11 and 12 refer to the diagram below, which shows two set-ups used for the investigation of gas exchange in germinating seeds:

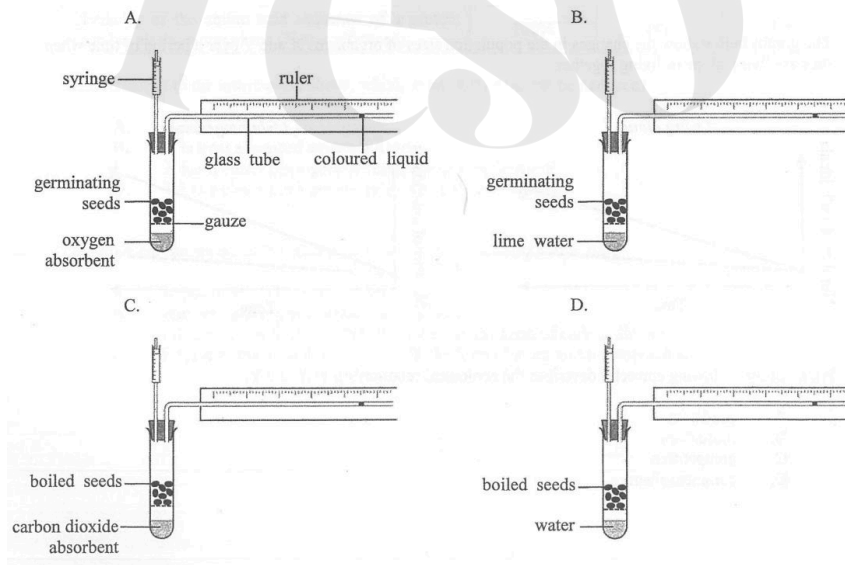


Assuming that the environmental conditions are the same, what will happen to the coloured liquid in each set-ups?

- | | Set-up P | Set-up Q |
|----|--------------------|--------------------|
| A. | moves to the right | moves to the left |
| B. | moves to the left | stays still |
| C. | stays still | moves to the left |
| D. | stays still | moves to the right |

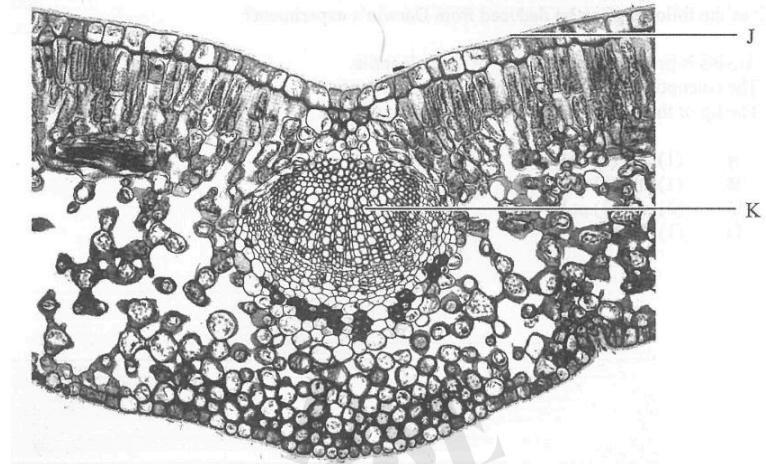
8. 2022Q12

Which of the following modified set-ups can be used to show the identity of the gas produced in the investigation?



9. 2022Q28

Question 28 refers to the photomicrograph below, which shows a section of a leaf:



Which of the following correctly describes the major function of layer J?

- It reduces water loss.
- It allows light to pass through.
- It protects the leaf from infection.
- It increases the photosynthetic rate.

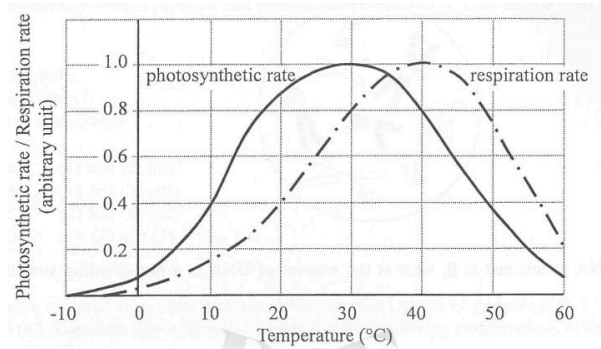
10. 2024Q1

Fertilisers usually contain nitrogen and phosphorous compounds, They can be used by plants for the synthesis of

- cell walls.
 - chlorophyll.
 - cell membranes.
- (1) and (2) only
 - (1) and (3) only
 - (2) and (3) only
 - (1), (2) and (3)

11. 2024Q8

Question 8 refers to the graph below, which shows the changes in the photosynthetic rate and respiration rate of a rice crop at different temperatures while the light intensity and concentration of carbon dioxide were kept constant:



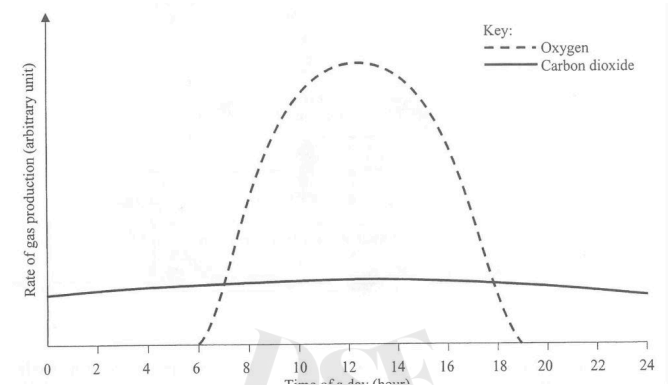
Which of the following interpretations about the graph is correct?

- A. The crop has maximum growth at 30°C.
- B. The crop does not absorb water at 40°C.
- C. The crop reaches compensation point at 35°C.
- D. The crop has greatest net production of food at 20°C.

LQ

1. 2012Q5

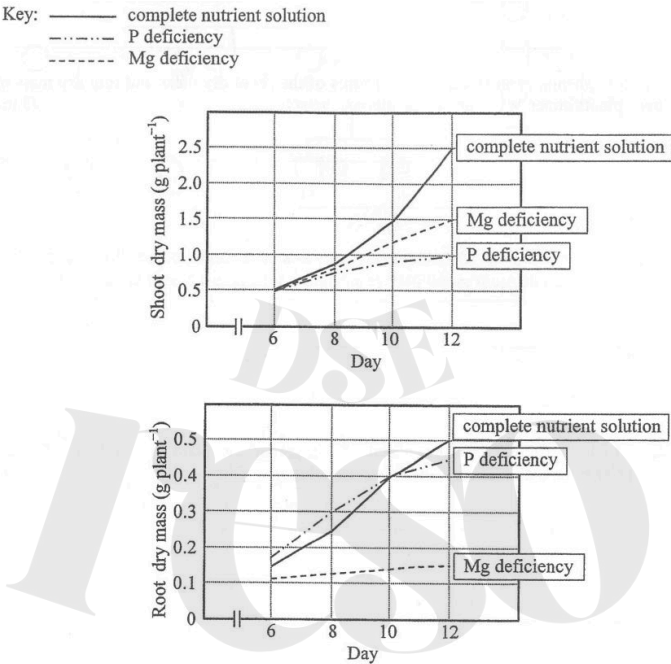
The graph below shows the oxygen production rate and carbon dioxide production rate of a local plant on a summer day:



- a) State the times at which there is no net exchange of gases into or out of the leaves. (1 mark)
- b) Sketch a line on the above graph to show the oxygen production rate of the plant on a winter day. (2 marks)
- c) The area below the line showing the oxygen production rate is usually greater than the area below the line showing the carbon dioxide production rate. Explain the importance of this observation. (4 marks)

2. 2016Q9b, c

To study the effect of mineral deficiency on shoot and root dry masses, bean plants were grown in a complete nutrient solution (a solution containing all essential nutrients for growth) or a nutrient solution without either phosphorus (P) or magnesium (Mg) for 12 days respectively. The dry masses of shoot and root were then measured. The results are shown in the graphs below.



- b) The leaves of the bean plants grown under Mg deficient conditions appeared yellow.
- i) Why did the leaves appear yellow? (1 mark)
 - ii) Use this phenomenon to explain the results of the shoot dry mass and root dry mass of the bean plants under Mg deficient conditions. (3 marks)
- c) i) Explain the difference in the overall dry mass of the plant grown under P deficient conditions and that in the complete nutrient solution. (2 marks)
- ii) It was hypothesized that P inhibits the export of photosynthetic products from leaves to roots. Use this hypothesis to explain the results of the shoot dry mass and root dry mass of the bean plants under P deficient conditions. (3 marks)

3. 2016Q11 - Essay

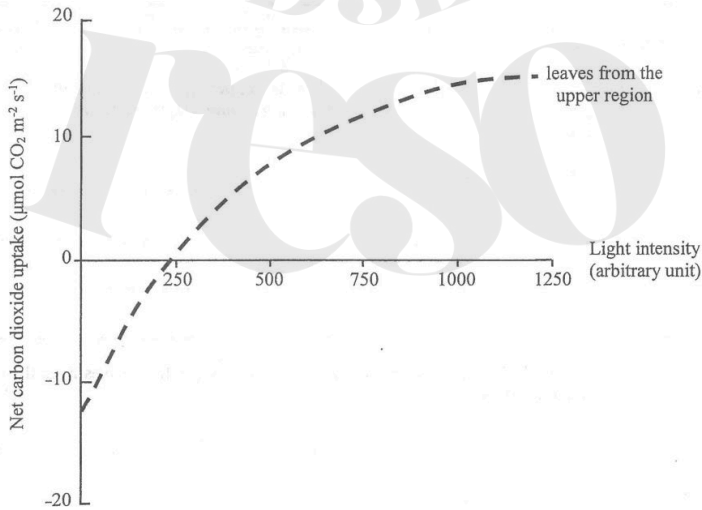
Gas exchange in organisms relies very much on diffusion. To be an effective organ for gas exchange, the leaves in plants and the lungs in humans share some common principles in their structural adaptations. Discuss how their structures are adapted to fulfill these common principles. Despite these similarities, explain why the operation of the breathing system in humans is more effective. (11 marks)

4. 2021Q8c

The table below shows the average blade area, blade thickness and thickness of the palisade mesophyll of leaves collected from the upper and lower regions of a tree species:

Location of leaves	Average blade area (cm ²)	Average blade thickness (μm)	Average thickness of palisade mesophyll (μm)
Upper region	62	177	45
Lower region	72	152	33

- c) Leaves at different regions of a tree are adapted to different light intensities. The graph below shows the change in the net carbon dioxide uptake by leaves from the upper region of a tree at different light intensities:



- i) Why are there negative values for net carbon dioxide uptake? (1 mark)
 - ii) On the above graph, sketch a line to show the change in net carbon dioxide uptake by leaves from the lower region of a tree at different light intensities. (2 marks)
- (Note: Neglect the difference in the average blade area between the two types of leaves when you sketch the line.)

Ans

2013Q3: C (72%)	2014Q6: A (70%)	2014Q7: A (74%)	2014Q8: B (47%)
2016Q9: A (79%)	2016Q10: C (67%)	2022Q11: B (51%)	2022Q12: B (74%)
2022Q28: A (85%)	2024Q1: C (48%)	2024Q8: D (28%)	

2012Q5

5	(a)	• 7:00 and 18:00 (1)	(1)
	(b)	• shorter light period (1)	(1)
		• overall rate lower (1)	(1)
	(c)	• the area below the line showing oxygen production rate represents the food production in 24 hrs (1)	(1)
		• whereas the area below the line showing carbon dioxide production rate represents the food consumption in 24hrs (1)	(1)
		• it is therefore important for food production to be greater than food consumption such that there is a net amount of food produced (1)	(1)
		• as a result, this provides energy for the plant to survive, grow and produce fruits(1)	(1)
			7 marks

2016Q9b, c

9	(b)	(i)	• without magnesium, chlorophyll cannot be formed (1), leaving the leaves yellow	(1)
	(b)	(ii)	• without chlorophyll, the rate of photosynthesis of the plant is lowered (1)	
			• as a result, there is not enough food produced for the growth of plants (1)	
			• therefore, both the shoot and root dry mass are smaller than that of the control / that grown in complete nutrient solution (1)	(3)
	(c)	(i)	• the overall dry mass of the plant under P deficient conditions is much smaller (1)	(2)
			• Because P is necessary for the formation of protein / nucleic acids / ATP (1) which are important for growth	
		(ii)	• without P, more photosynthesis product is transported from leaves to root (1)	
			• as a result, the shoot dry mass is a lot lower than that of control (1)	(3)
			• while the root dry mass maintains more or less the same (1)	
				9 marks

2016Q11

11	Adaptations to be an effective organ for gas exchange (A) max.=5			(max. 5)
		• large surface area for diffusion of gases (1)		
		❖ numerous air sacs in the lungs of humans vs spongy mesophyll with numerous air spaces in leaves / numerous leaves in plants (1)		
		• a moist surface for dissolving of gas (1)		
		❖ presence of a water film on the inner surface of the air sac vs that of the surface of spongy mesophyll (1)		
		• short diffusion distance for exchange of gas between internal and external environment (1)		
		❖ one-cell thick wall of air sacs and capillary versus flat and thin leaves(1)		
	Human lungs are considered more effective in gas exchange (S) max.=3			(max. 3)
		❖ there is active ventilation in humans, breathing movements draw in and expel air actively (1)		
		❖ oxygen diffused in are transported away by the blood of the capillary network surrounding the air sacs (1)		
		❖ both of the above maintain a steep concentration gradient for diffusion of gases (1)		
		❖ while leaves rely on passive ventilation / diffusion only (1)		
	Effective communication (C)			(max. 3) 11 marks

2021Q8c

8	(c)	(i)	• respiration is faster than photosynthesis (1), resulting in a net release of carbon dioxide	(1)
		(ii)	• line with a higher starting point (1) (data interpretation: the amount of respiring tissue is smaller, so the amount of carbon dioxide produced is smaller when light intensity is zero)	
			• line indicates that a lower net CO ₂ uptake can be reached (1) (data interpretation: the amount of photosynthetic tissue is smaller, therefore the amount of carbon dioxide uptake is smaller)	(2)
				3 marks